Beyond a Single Error Rate: Confusion-Pair Diagnosis of Cross-Generation Tamil OCR Failure

R.Bhuvaneswari
Associate Professor
Computer Science and Engineering
Rajalakshmi Engineering College
Chennai, India
bhuvanacheran@gmail.com

Varsha Thomas
Computer Science and Engineering
Rajalakshmi Engineering College
Chennai, India
230701372@rajalakshmi.edu.in

Uma AL
Computer Science and Engineering
Rajalakshmi Engineering College
Chennai, India
230701368@rajalakshmi.edu.in

Varsha H
Computer Science and Engineering
Rajalakshmi Engineering College
Chennai, India
230701371@rajalakshmi.edu.in

Tanisha C A
Computer Science and Engineering
Rajalakshmi Engineering College
Chennai, India
230701390@rajalakshmi.edu.in

Abstract— ***A benchmarking paper [16], which shows that grapheme-cluster-aware CER/WER, rather than codepoint-level CER, is the proper metric to use for scoring Tamil OCR systems, documents a leaderboard of seven systems across three generations on a stratified, real-world dataset. However, an aggregate error rate gives no information as to why a system makes mistakes – whether it outputs an erroneous but plausible reading, a visually similar misreading akshara, or something entirely unrelated to the original – three different types of error with entirely different implications for downstream pipelines. In this paper, we operationalize the reorder/misread F1 decomposition used in the companion paper into a five-part taxonomy of substitution edit confusion pairs,***

recited/ambiguous/misreading/hallucination/unrelated,

referred to herein as the Crutch Ratio, and apply it to 369 scored substitution errors from the seven baseline systems on four of the five benchmark corpora documented in the companion paper. The most strikingly uniform conclusion is the negative finding that misreading confusion pair category – genuine visual confusion of one akshara for another – is entirely absent for each of the seven engines considered. Rather, errors decompose into two starkly distinct behaviors: an erroneous yet plausible akshara sequence, dominant for Chandra, or total structural breakdown to fluent but unrelated text, dominant for EasyOCR, PaddleOCR-VL, and Surya. Three additional examples of system-specific error cases – the bimodal layout gate driven failure mode of Surya, the silent multimodal collapse to Tibetan text and digits in PaddleOCR-VL, and the English word hallucinations of Gemini on a synthetic example – demonstrate clearly why an aggregate CER score alone fails to account for failure modes that need separate solutions. We conclude by defining, as the next step in developing this framework of analysis, an akshara level Structural Integrity Score (ASIS) and Decomposed Error Profile (DAEP), along with a cross-system error correlation analysis – both of which we define with formulas in this paper, but which we have chosen, in the interest of avoiding unverified numbers, not to include in this paper's results.

Index Terms—Tamil OCR, error identification, confusion set classification, hallucination identification, akshara level structure analysis, across generation OCR, low resource script OCR

I. INTRODUCTION

In a companion study [16], it is asserted that Tamil OCR evaluations should evolve from codepoint-level CER to grapheme-cluster-sensitive CER/WER, with a leaderboard for seven representative OCR baselines, from classical, open vision-language models, to frontier vision-language models, based on a real-world Tamil document collection with stratified samples, at pilot-level (n=100, IndicSTR12) and corpus-level (n=644, five datasets). This leaderboard gives an account of how well or poorly the respective OCR engine performs. This leaderboard, however, does not give any information on why: two OCR engines might have similar aggregate CER levels but fail completely differently, and in completely different ways. The OCR engine that produces reasonable but incorrect text could be corrected by a spell checker or language model, but one that produces fluent but totally irrelevant text could mislead its users.

Such differentiation is especially significant for Tamil since it is possible in theory to extend the decomposition introduced in the accompanying paper to distinguish harmless artifacts resulting from encoding order from actual misrecognitions (its Eq. 3) – the latter could be, respectively, either a visually-based misinterpretation of one akshara for another, a linguistically-based substitution of one valid word for another, or something more extreme, having no visual nor linguistic connection to the source character whatsoever. Differentiating between these is impossible based solely on CER statistics, which, to the best of our knowledge, has not been attempted with Tamil OCR before in Section II literature.

Specifically, this paper provides four new contributions. Firstly, we provide an operationalization of the F1 reorder/misread decomposition in terms of a five-category confusion-pair taxonomy, which we term the Crutch Ratio, categorizing every substitution edit generated by the seven engines studied in our companion paper into one of the categories recite, ambiguous, misread, hallucination, and unrelated. Secondly, we provide this taxonomy for the full corpus (369 scored substitutions, four corpora, seven engines), providing a negative result, namely the complete lack of the misread category across all seven engines. Thirdly, we present three case studies of particular engines — Surya, PaddleOCR-VL, and Gemini — detailing a particular failure mode of each of those engines, invisible in a simple CER score. Finally, we propose an exact specification for an akshara-level Structural Integrity Score (ASIS), a Decomposed Error Profile (DAEP), and an error-correlation statistic across systems, which constitute the next logical step in the diagnostic methodology; in keeping with the evidentiary standard that has been consistently upheld throughout this project, we provide only the specifications of this methodology but no corpus-wide results, since neither has been validated against a hand-adjudicated pilot study as yet.

The rest of this paper is structured as follows. In Section II, we will discuss relevant literature. In Section III, we introduce the Crutch Ratio taxonomy along with the ASIS/DAEP and cross-system extensions that were specified but not yet executed. In Section IV, we describe the implemented classification pipeline. In Section V, we analyze the confusion-pair classification and three specific-case studies of each engine.

II. RELATED WORK

Previous work in early Tamil OCR [11],[12] and uTHCD dataset for character isolation [2] have shown feasibility of recognition and availability of data, but none of which provides a failure analysis other than reporting the overall accuracy. IndicSTR12 [4] and IndicDLP [3] datasets are realistic and large-scale but they fail to analyze what causes the failure of a particular recognition system on a given input. The works related to Tesseract legacy font adaptation [9] and comparative studies for Sinhala and Tamil [8],[10] report differences in accuracy but end up with code-point level CER.

Multilingual/general-purpose OCR/VLM architectures such as PaddleOCR-VL [6] and E-ARMOR [7] provide natural baselines to consider, yet none have released a Tamil-specific confusion-pair-level error analysis as proposed here. In terms of multimodal model evaluation, OCRBench [13] and OCRBench v2 [14] laid down the benchmarking foundation yet favor predominantly Latin/Chinese material and do not provide any Tamil-specific tools for model evaluation. The GlotOCR Bench [5] also validates this claim of models' struggle "beyond a handful of Unicode scripts" including Tamil yet uses a codepoint-level evaluation framework and not an akshara-level one as proposed in this paper.

Grapheme Kit [1], an adaptation of CER/WER for Unicode graphemes, is the best existing approximation of the metric layer that the current paper is founded upon, though this tool does not generalize beyond grapheme pair taxonomies and akshara level structural diagnosis. The body of literature demonstrates that there are existing tools for grapheme based evaluation but, to our knowledge, have not yet generalized to a confusion pair taxonomy, engine-specific analysis of error modes, or akshara level structural diagnosis for Tamil.

III. DIAGNOSTIC FRAMEWORK

A. SHARED CORPUS AND BASELINE RECAP

This work uses the stratified corpus, the seven baselines for OCR (Tesseract, EasyOCR, Surya, PaddleOCR-VL, Gemini 3.5 Flash-Lite, Gemma 4 26B, and Chandra), and the machinery of alignment of graphemes and clusters as described in our companion paper [16], unchanged. All notation used below (N, R, M, edit set E) is taken from the latter.

B. THE CRUTCH RATIO: A FIVE-CATEGORY CONFUSION-PAIR TAXONOMY

F1 decomposition from the companion paper (given by Eq. 3) classifies each of the edit operations either as reordering error or as misreading error. This paper makes further classifications under the category of misreading error for each of the substitution operations, while ignoring deletions and insertions as there are no confusion pairs associated with them. Each substitution edit operation between the hypothesis and the reference graphemes is classified into one of:

- recitation – an incorrect but reasonable-seeming akshara string (a linguistic replacement).
- ambiguous – a valid but double-meaningful string both visually and linguistically.
- misreading – a perceptual error involving similarly-looking aksharas but linguistically implausible.
- hallucination – fluent speech without any connection with the source string.
- unrelated – writing not having anything to do with the source string at all.

The first three categories are grouped, following the companion paper's naming, as reciting/ambiguous/misreading; the last two are jointly reported as structural breakdown, since both indicate the system has stopped attempting to transcribe the source content at all.

C. SPECIFIED EXTENSION I — AKSHARA-LEVEL ASIS/DAEP

While the Crutch Ratio labels substitution editing of each individual unit, ASIS and DAEP label each entire akshara unit, matching them at the akshara level (more stringent than the grapheme-cluster level mentioned earlier, as the Tamil conjunct கஷ breaks up into two grapheme clusters but only forms one akshara). Let A_total represent the total number of reference aksharas, while A_intact represent the number not edited by any change at the akshara level:

$$\text{ASIS} = A_{\text{intact}} / A_{\text{total}}$$

Each non-intact akshara belongs to one of only four types: base only, matra only, ordering or attachment, or compound; and DAEP computes the proportion of each type in A_total:

$$\text{DAEP}_{\text{base}} + \text{DAEP}_{\text{matra}} + \text{DAEP}_{\text{attach}} + \text{DAEP}_{\text{compound}} = 1 - \text{ASIS}$$

Status: fully specified, validated by worked example (reference அகழரம்: 3 out of 4 aksharas present, ASIS = 75.0%, the single missing akshara labeled as base only, DAEPbase = 25.0% — the identity tests: 25.0% = 1 – 75.0%), but yet to be applied on the corpus level. It needs a separate akshara segmenter different from the grapheme cluster segmenter employed in the case of the Crutch Ratio, as well as a pilot of 50 to 100 lines hand-checked for category labeling consistency before any corpus-level number is quoted — as all other extension tier metrics in this project.

D. SPECIFIED EXTENSION II — CROSS-SYSTEM ERROR CORRELATION

Regardless of whether the above confusion-pair distribution indicates that each engine fails independently on structurally difficult inputs or there are multiple engines that share a similar failure mode is a different issue from classification. For each pair of engines (e1, e2), binarize the scored lines based on whether they are right/wrong for CER = 0, and then calculate the Matthews correlation coefficient based on the 2 × 2 outcome matrix (n_{11}, n_{10}, n_{01}, n_{00}):

$$\varphi(e_1, e_2) = \frac{n_{11}n_{00} - n_{10}n_{01}}{\sqrt{(n_{11}+n_{10})(n_{11}+n_{01})(n_{00}+n_{10})(n_{00}+n_{01})}}$$

STATUS: fully defined and tested on a previous pilot corpus (only IndicSTR12, n=300), where all pairwise φ's were in a weak-to-moderate range between 0.05 and 0.45 – a real but small signal, not a large one. This test predated the larger five dataset corpus (n=644) used elsewhere in this paper, so it is mentioned here merely as methodological precedent, not re-stated as a full corpus result; computing φ on the new corpus was listed as future work (Section VI).

IV. IMPLEMENTATION

A. CORPUS AND BASELINE EXECUTION (RECAP)

Construction of corpus, stratification marking, and the seven engines execution are identical to that of the companion paper [16]; this paper uses the very same scored results and not any engines.

B. CRUTCH RATIO CLASSIFICATION PIPELINE

Each substitution edit generated by the grapheme-based alignment (Section IV-C in the accompanying paper) is subjected to the five-category classification scheme described in Section III-B. Categories of recite/ambiguous/misread are determined by a rule-based layer that tests linguistic and visual plausibility relative to a reference confusion-pair dictionary, while hallucination/unrelated are determined using the script/fluency classifier on the entire line prediction, followed by manual spot-checking— for instance, each of the eleven classified outputs as latin_english in the synthetic IndicSTR dataset (Section V-B) was validated individually against the Tamil ground truth prior to reporting as hallucinations.

C. FAILURE-MODE DIAGNOSTIC TOOLING

Two additional detectors specific to engines are present along with the Crutch Ratio. The Layout-Gate detector examines the raw JSON output of Surya and identifies the difference between the engine which tried and failed to read the text and the one whose layout stage did not provide a crop for the text recognition in the first place (V-B). The script-identification classifier assigns a script label to all non-Tamil output texts in order not to have a wrong, but confident transcription identified as the right one (V-B). Two more artifacts have been discovered, fixed, or declared dead in the process: the literal "empty response" string of Gemma output five times was caught by the artifact classifier, and Chandra's markdown-image-caption hallucinations discovered in the older, superseded output directory were determined to be present only there.

V. RESULTS

A. CRUTCH RATIO: CONFUSION-PAIR QUADRANT BREAKDOWN (N = 369)

Confusion pair analysis is presented in Table III and Fig. 5, aggregated across all four test corpora on which the classification was performed (indicstr12_sample is not included in this pipeline). One clear finding is that the quadrant corresponding to visual confusion of aksharas is completely empty in every one of the seven engines. For an engine in which confusion between visually similar aksharas is an a priori possible failure mode, this is a meaningful negative finding – none of the seven engines being evaluated makes mistakes through misreading aksharas in a manner which is visually similar to each other. Instead, the errors can be cleanly split into two very different modes of failure – a likely but incorrect akshara sequence (reciting, dominant in Chandra 37 of 46 substitutions, 80.4%), or a complete breakdown into fluent but unrelated/hallucinated text (dominant in EasyOCR, PaddleOCR-VL, and Surya, > 70%). This difference is functionally significant – the first category is a mistake that can be fixed by a downstream spell-checking system/LM, while the latter is active misinformation.

Engine	Recit.	Ambig.	Misread	Hall.	Unrel.	n
Tesseract	11	1	0	51	2	65
Surya	12	0	0	28	0	40
PaddleOCR-VL	3	0	0	36	0	39
Gemini	20	6	0	21	2	49
Gemma	10	9	0	17	0	36
Chandra	37	5	0	3	1	46
EasyOCR	9	4	0	78	3	94

TABLE III — Crutch Ratio Quadrant Breakdown by Engine (pooled, 4 datasets, n = 369 scored substitutions)

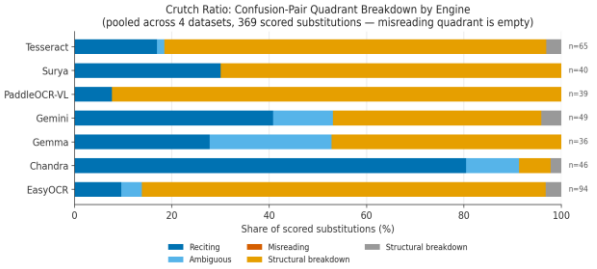


Fig. 5. Crutch Ratio confusion-pair quadrant breakdown by engine, pooled across four datasets (369 scored substitutions). Note: the underlying chart's legend currently double-labels the hallucination and unrelated segments both as "structural breakdown" and assigns misreading an unused swatch (misreading = 0 for every engine, per Table III) — a labeling fix, not a data error, flagged here for correction before this figure is finalized.

B. ENGINE-SPECIFIC DIAGNOSTIC CASE STUDIES

In addition to the general Crutch Ratio decomposition, three engine-level results are discussed here, as all of these would not be seen or indeed misinterpreted using only one CER figure.

Surya's failure is bimodal by dataset, not a blanket language weakness. Surya's document-layout classifier labels small, isolated word crops Picture and skips OCR on them entirely, confirmed by inspecting its raw output JSON and by forcing a Text label, which made it caption the photo instead of reading it. Table IV shows this is a crop-format/layout-gate failure, not a uniform language weakness: Surya returns an empty string on 100% of hw_words_tamil_sample crops but only 1% of synthetic_r_tamil_sample crops.

Dataset	Empty	Correct Tamil
hw_words_tamil_sample	100%	0%
indicstr12_sample	93.3%	4.7%
synthetic_indicstr_tamil_sample	69%	17%
synthetic_r_tamil_sample	1%	95%
self_collected_handwritten	0%	68.2%

TABLE IV — Surya Empty-Output Rate by Dataset

PaddleOCR-VL's silent multi-script collapse needs the caveat more than Surya does. Unlike Surya, PaddleOCR-VL never returns an empty string — every failure looks like a confident, complete transcription. Of 271 non-Tamil outputs classified across the benchmark, the two dominant collapse modes are Tibetan script (95, ~35%) and pure digits (74, ~27%), with the remainder spread across Telugu, Arabic, CJK, Thai, Hangul, and Latin. A raw CER/WER number cannot distinguish a correct transcription from confident Tibetan.

The 11% bucket that contains the Latin/English errors in Gemini is a real hallucination and not an artifact of evaluation. This means that all eleven outputs that were classified as latin_english have been manually verified against their Tamil truth, and they are fluent English words and phrases (e.g., “SWALLOW”, “Gatorade”, “mcafee”) that do not have any relationship with the source document, whether semantically or visually. In the same test set, the only engine out of the seven to commit a substitutive error is PaddleOCR-VL (n = 8, 100% structural breakdown).

VI. FUTURE WORK

- Carry out the ASIS/DAEP akshara-level attribution for Section III-C at the level of the whole corpus, behind its own hand-adjudicated pilot and inter-annotator agreement check, rather than reporting an unvalidated number.
- Recalculate the cross-system error correlation statistic (φ, Section III-D) using the new five-dataset, n = 644 corpus; the 0.05–0.45 range reported from the previous n = 300, single-dataset pilot is not current until this is done.
- Adjudicate the Crutch Ratio's pilot inter-annotator disagreement throughout the corpus, so that the taxonomy gains robustness through more than one pilot check.
- Correct the Fig. 5 legend label error pointed out in Section V-A prior to presenting this diagnostic framework anywhere else.
- Considering the repeated use of Tamil alongside structurally analogous Sinhala abugida in the literature [8]–[10], the Crutch Ratio taxonomy is expected to be adaptable with slight modifications to other Brahmic scripts..

VII. CONCLUSION

The current paper applied the seven engines and corpus-wide benchmark presented in a companion paper [16], but addressed a question that benchmark did not consider: not how much error each system produced, but the nature of error it made. Applying the five category Crutch Ratio taxonomy based on F1 reorder/misread decomposition to the 369 substitution errors found in the benchmark exposed a true negative result — visual misreading between similarly shaped aksharas never happens — while differentiating linguistic guessing from structural failures, two different failure types requiring different fixes. The three engine-specific studies also indicated that Surya's failure was a layout gate artifact rather than any linguistic limitation, that the silent multi-script failure of PaddleOCR-VL is completely undetected by CER/WER, and that Gemini's hallucinations were real and not a classifier noise. Full specification of structural attribution at the akshara level (ASIS/DAEP) and corpus current cross-system correlation analysis are outlined here as the next stage in the development of this framework, and are reserved, purposefully, for validated future research.

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